03/05

Wednesday, 5 March 2025 9:32 AM

Midterm 03/07 !

→ Open book ; open notes ; 10 questions ; very picky (highlight all points)
(No internet)

→ Conceptual questions

Finite Volume Method

FD → b
↳ generalized coordinate
(mapping from cartesian to via jacobian

Euler equation : $\frac{\partial U}{\partial t} + \nabla \cdot F = 0$;

$F = fi + gj$

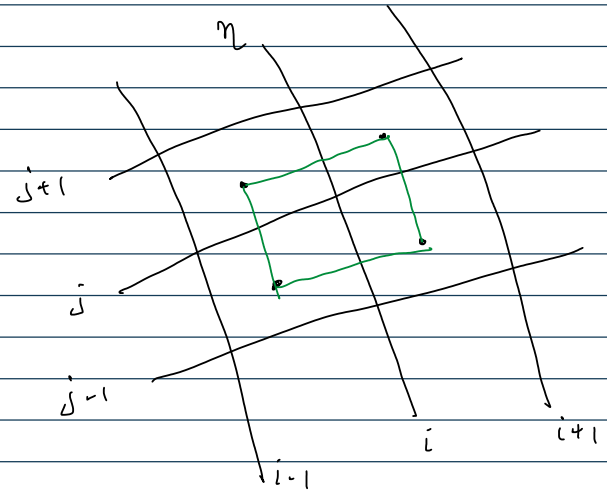
→ Still need to generate grid but do not have to map to generalized coordinate

→ Integrate over CV and use Gauss divergence theorem $\int_V (\nabla \cdot F)\, dV = \int_A F \cdot dA$

"What goes in, comes out"

$$V \frac{dU}{dt} + \int_A F \cdot dA = 0$$

→ If CV is moving → need Leibniz's rule



Flux . surface area

→ then choose discretized volume method

↓

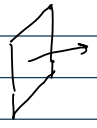
Backward, backward bias

Cell central method vs. vertex centered

How to calculate area of each face of a 3D cell ?

vectors on surface





$\vec{S}_1 = \frac{1}{2}(r_1 \times r_2)$

cell volume = $\frac{1}{3} r \left(S_1 + S_2 + S_2 \right)$

Converting FDM to FVM

1) Transform GE in generalized coord in strong conservation-law form

2) :

3) discretize PDE

Potential flow problem:

Potential flow is used to model irrotational where the viscosity effects are assumed to be negligible - typically externally flow over solid surfaces or flow at high Reynolds number. In such cases, the thickness of the boundary layer is assumed to be much smaller than the dimensions of the object.

The given equation is the Laplace equation using stream function. Stream function (ψ) can be defined for any 2D flow such that the stream function is constant along a streamline. Additionally, change in stream function between a pair of streamlines equals the volummetric flowrate between the two streamlines. Finally, lines of constant streamlines are perpendicular to the lines of constant velocity potential (Φ).